FemboyEngine2D API Reference Manual

C++26 data-oriented engine - complete public surface, contracts and a build-from-zero walkthrough

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1. Scope, conformance and engine rules

This manual specifies the public surface of FemboyEngine2D: every header under Source/Engine and the editor infrastructure that a host links against. It is written against the release-candidate tree, so a symbol described here exists in the headers that ship with that tree, and the appendix index is generated from those headers on every run.

The engine targets C++26 with GCC 16 or newer and Clang 19 or newer; the Windows release candidate is built with clang-cl against the MSVC standard library. Four rules are load bearing and every public API is designed around them:

- Simulation state is plain, trivially copyable data. No virtual calls, no owning pointers and no std::string belong in a component, snapshot or network payload.

- Hot paths do not allocate. Frame scratch memory comes from FrameArena, logs go through fixed char buffers, and callbacks are function pointers plus void*, never std::function.
- Determinism is a property of the data, not of the scheduler. Every subsystem exposes a checksummed image, and NetRollbackSession can only replay what it can reproduce byte for byte.
- No RTTI in engine code: dynamic_cast and typeid are rejected by the style gate. Cast<> walks stable type ids instead (see section 3).

> The style gate (Source/Tools/Dev/CheckStyle.py) enforces the naming and structure rules: PascalCase file stems, Allman braces, braced control bodies, In-prefixed parameters, Scoped-prefixed locals, no trailing line comments and no Cyrillic in engine sources.

2. Building, targets and compiler flags

One CMake tree produces three kinds of artefact: the engine static library (FemboyEngine2D::Engine), the editor library and application (FemboyEngine2D::Editor), and the offline tools. A shipped game links the engine only; Dear ImGui, ImGuizmo, ImPlot and the TOML writer never enter a shipping target.

CMake option	Default	Effect
FBY_BUILD_EDITOR	ON	Builds fby_editor, the editor app and the panel plugins.
FBY_BUILD_TESTS	OFF	Builds the ctest suite under Source/Tests.
FBY_BUILD_TOOLS	ON	FempakCooker, LocTool, SceneTool, ShaderCompiler, AotCompiler.
FBY_BUILD_PLATFORM_PROVIDERS	ON	Steam and Epic Games Store provider plugins.
FBY_RHI_BACKEND	auto	auto, vulkan, metal or null. auto selects per platform.

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo \
      -DFBY_BUILD_TESTS=ON -DFBY_BUILD_EDITOR=ON
cmake --build build --target fby_editor
ctest --test-dir build --output-on-failure
python Source/Tools/Dev/CheckStyle.py
```

The compiler wrapper (cmake/FbyCompiler.cmake) owns the C++26 flag, the warning set and the sanitizer switches. Do not add flags per target: a target that needs a different dialect is a target that should not exist.

3. Core value types, FORCEINLINE and the cast family

Definitions.hpp is the contract header of the engine. It supplies the export macros, the assertion macros (FBY_ASSERT, FBY_VERIFY, FBY_UNREACHABLE), the stable TypeId machinery, the cast family and the component registration macros. Every other header includes it, directly or transitively.

```
#if defined(_MSC_VER)
```

```
#define FORCEINLINE __forceinline
#elif defined(__GNUC__) || defined(__clang__)
#define FORCEINLINE inline __attribute__((always_inline))
#else
#define FORCEINLINE inline
#endif
```

FORCEINLINE is the only inlining spelling in public headers. It is used for accessors, hot arithmetic and template casts; it never replaces a measured optimisation. The three families are:

Symbol	Contract
FemCast<To>(value)	Unchecked static_cast with reference collapsing. Use when the concrete type is proven by construction (factory output, storage tag). No runtime cost.
Cast<To>(object)	Checked cast. Upcast is a compile-time static_cast; downcast compares the object's GetTypeId() against StableTypeId<To>() and its registered parent chain; unrelated types fail to compile. Returns nullptr on a mismatch, never throws, never uses RTTI.
CastComponent<To>(registry, entity)	ECS spelling of Cast<>: EnTT proves the component type at compile time and the registry validates the entity. Returns nullptr when the entity lacks the component.

```
// Downcast through the stable type table, not through RTTI.
Fby::TypeObject* base = factory.Create("Sprite");
if (Fby::Sprite* sprite = Fby::Cast<Fby::Sprite>(base))
{
    sprite->Animate(dt);
}

// Registration binds Sprite to TypeObject and derives its stable id.
FBY_REGISTER_TYPE(Fby::Sprite, Fby::TypeObject);
```

StableTypeId<T>() is a consteval FNV-1a of the type's registered name (or the compiler signature when no name is pinned). A type that changes its registered name changes its id, which is why component names are part of the serialized schema and why Snapshot v4 stores a LayoutHash next to the name.

4. FrameArena: the only frame allocator

FrameArena is a bump pointer over caller-owned storage. It is reset at the start of a frame and never frees individual allocations. Systems that need transient arrays take it from SystemContext::frameArena; a system that calls operator new in its frame path is a defect, not a style preference.

```
Fby::FrameArena arena(storage);          // caller owns storage
arena.Reset();                           // once per frame
auto* items = arena.AllocateArray<MeshRenderItem>(count);
if (items == nullptr)
{
```

```
    // Arena exhausted: degrade, never allocate on the heap.
    return;
}
const std::size_t used = arena.Used();
const std::size_t capacity = arena.Capacity();
```

- Allocate() returns nullptr instead of throwing when the arena is exhausted; callers must handle the failure.
- AllocateArray<T>() static_asserts std::is_trivially_copyable_v<T>: the arena stores POD payloads only.
- Over-aligned types must pass their alignment explicitly; the default is alignof(std::max_align_t).

5. JobSystem and per-thread diagnostics

JobSystem owns a pool of std::jthread workers. Tasks are function pointers plus a context pointer plus a half-open range, so submitting a task never allocates. ParallelFor splits a range into chunks, lets every participant pull the next chunk from an atomic counter, and always runs at least one chunk on the caller, which keeps single-core machines correct and worker re-entry safe.

```
jobs.ParallelFor(entityCount, 4096, [&](std::uint32_t begin, std::uint32_t end)
{
    for (std::uint32_t i = begin; i < end; ++i)
    {
        Integrate(entities[i], dt);
    }
});
jobs.WaitIdle();

// Diagnostics: one row per thread, slot 0 is the main thread.
std::array<Fby::JobSystem::ThreadLoad, Fby::JobSystem::kMaxThreadSlots> loads{};
const std::uint32_t slots = jobs.SampleThreadLoad(loads);
```

ThreadLoad carries busyNanoseconds (wall time inside engine work), taskCount and a 16-byte name. The counters are one cache line per thread, so two workers finishing at the same time do not contend. SceneDiagnosticsPanel turns two samples into a per-thread occupancy percentage.

> ThreadSlotCount() is ThreadCount() + 1 and is clamped by the fixed MCounters array; a device with more than 64 workers reports the first 64 slots.

6. ECS, scheduling and snapshots

The registry is EnTT. Systems are registered on a SystemGraph in one of four phases (PreUpdate, FixedUpdate, PostUpdate, Render). SystemContext is the only parameter a system receives; it carries the registry and job system directly and groups the subsystem pointers into domain bundles (render, assets, audio, physics, text, accessibility, gameplay), so a system reads only the bundles it declares. Scheduled systems take a narrow view: `PhysicsSystemView`, `RenderSystemView`, `GameplaySystemView`, `AudioSystemView`,

`UiSystemView`, `AccessibilitySystemView` or `CoreSystemView`, adapted by `Bind*System<&Function>()`.

```
Fby::Ecs::SystemGraph graph;
graph.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem, &physics);
graph.Add(Fby::Ecs::Phase::FixedUpdate, "movement", &MovementSystem, nullptr, 1ull <<
0);
graph.Add(Fby::Ecs::Phase::PostUpdate, "platform", &GogAchievementSystem, &gog);

Fby::Ecs::SystemContext context{registry, jobs};
context.physics = &physics;
context.window = &window;
graph.Build();
graph.Run(Fby::Ecs::Phase::FixedUpdate, context);
```

Snapshot v4 is a canonical, checksummed byte image of the registry. Components are written by hand-written codecs when they contain runtime handles (b2BodyId, b2ShapeId); the snapshot never serializes those handles, and the restore path recreates bodies from the component fields. The schema stores the Mirror name and LayoutHash of every component, so a field change is rejected instead of silently misread.

API	Contract
GetOrAdd<T>(registry, entity)	Adds the component when missing and returns a stable reference.
ecs::SnapshotWriter / SnapshotReader	Writes and validates the canonical image; checksum is FNV-1a over the payload.
Fby::Net::NetRollbackSession	Consumes snapshot v4 and replays ticks on late input (section 10).

7. RHI, render items and frame statistics

The RHI is a hand-written Vulkan/Metal layer. Frames are recorded into persistent ring buffers; sprite items are 48-byte PODs; meshes are drawn through one indirect call per material/layer; linear-blend skinning is one compute dispatch. FrameStats exposes what the pass actually submitted, which is what SceneDiagnosticsPanel plots.

FrameStats field	Meaning
items / batches / drawCalls	Sprite items, batches and submitted draws.
meshItems / meshIndirectCalls	MeshRenderItem count and indirect draws.
skinDispatches / skinRanges	Skinning compute dispatches and ranges.
uploadedBytes	Bytes uploaded this frame.

> The editor's Viewport renders into an offscreen target, not into the swapchain. EnsureViewportTarget() sizes that target in physical pixels; a target sized in points is the usual cause of a blurry Viewport on a HiDPI monitor.

8. Materials: cooked graphs, the pipeline cache and the shipped path

Material authoring and material compilation live on opposite sides of the engine boundary. `Render::MaterialGraph` is the authoring DAG: fixed capacities (128 nodes, 384 pins, 64

parameters), trivially copyable, and hashed without the editor positions. `Render::MaterialGraphCompiler` turns a validated graph into GLSL 450 text. `fby_matc` runs that compiler and glslang at build time and writes a `.femmat` payload, so a shipped game never links glslang and never parses GLSL. The full rationale is in [ADR 0002](../../adr/0002-materials-in-engine.md).

Piece	Header	Contract
`Render::MaterialGraph`	`Engine/RHI/MaterialGraph.hpp`	Authoring DAG; `Hash()` is the pipeline identity and excludes UI positions
`MaterialGraphCompiler`	`Engine/RHI/MaterialGraphCompiler.hpp`	DAG to GLSL 450 and the shader contract flags
`EncodeMaterialGraph` / `DecodeMaterialGraph`	`Engine/RHI/MaterialAsset.hpp`	`.fgraph` wire format: explicit little-endian fields, no struct memcpy
`EncodeCookedMaterial` / `DecodeCookedMaterial`	`Engine/RHI/MaterialAsset.hpp`	`.femmat` payload: 32-byte header, vertex and fragment SPIR-V, zero-copy decoding
`Render::MaterialPipelineCache`	`Engine/RHI/MaterialPipelineCache.hpp`	128 pipelines keyed by graph hash, reference counted, LRU eviction, no heap use
`Rhi::RHI::SupportsRuntimeShaderPipelines`	`Engine/RHI/RHI.hpp`	Capability query: Vulkan returns true, Null and Metal return false

```
# Author the graph once, cook it in the build step
build/bin/fby_matc assets/materials/tinted.fgraph -o assets/materials/tinted.femmat

// `InBytes` is the .femmat payload from the VFS or a package entry.
Render::CookedMaterialView ScopedMaterial;
if (!Render::DecodeCookedMaterial(InBytes, &ScopedMaterial))
{
    return false;
}

// One cache per device: acquire at load time, release at shutdown.
const Rhi::ShaderPipelineHandle ScopedPipeline = InMaterials.Acquire(InRhi,
ScopedMaterial);

Rhi::MeshRenderItem ScopedItem;
ScopedItem.materialPipeline =
    ScopedPipeline.IsValid() ? ScopedPipeline.id : Rhi::kDefaultMaterialPipeline;
```

- The graph hash is the pipeline identity. Editing a graph produces a new hash, so the old pipeline becomes an unreferenced cache entry instead of being mutated in place.
- The contract mask travels with the payload: generated vertex stage, fragment view projection, material push constants, world and vertex position varyings. A live entry whose hash arrives with another contract is refused, because that is a different shader.
- `Acquire()` adds a reference and `Release()` drops one. A pipeline at zero references stays resident, so a material acquired once per frame never recreates a pipeline; it is destroyed by LRU eviction, `ReleaseAll()` or `Shutdown()`.

- The cache never allocates and never formats text on the frame path; `LastError()` returns the fixed error buffer.
- Backends without runtime shader pipelines keep the engine shaders. Query `SupportsRuntimeShaderPipelines()` instead of probing for an invalid handle.

Payload	Fields	Validated on load
`.fgraph`	magic `FBYG`, version, node/pin/param counts, nodes, pins, params	magic, version, counts against capacity, exact size, node kinds, pin and param indices
`.femmat`	magic `FBYM`, version, contract mask, graph hash, vertex and fragment byte lengths	magic, version, size against the lengths, both SPIR-V modules, contract consistency

The editor keeps live editing on top of the same cache: `Fby::Editor::MaterialGraphRuntime` recompiles a graph when its hash changes, reports per-tab errors and delegates pipeline ownership to `MaterialPipelineCache`. `MaterialGraphRuntime::SetDomain` switches the live compile between the mesh and the sprite contract, and the panel's `SaveGraph`/`LoadGraph` write and read the `.fgraph` payload the cooker consumes. A shipped build skips glslang entirely and only calls `DecodeCookedMaterial` and `Acquire`.

Sprites take materials through the third artefact: `MaterialCompileOptions::domain` selects the sprite domain, where the generated fragment stage matches the `quad.vert` varyings (`vUv`, `vColor`, `vLayer`, and `vWorld` when a node needs it) and samples the item's atlas slice through `vLayer`. `Time` is available: the sprite push block carries a material clock at offset 80, so the generated stage reads `pc.materialParams.x` exactly like a mesh shader. `fby_matc --sprite` cooks that variant. `DrawOrder::MaterialBucket` places the sprite in a bucket, `ChangeMaskRenderExtract::SetMaterialPipelines` maps buckets to cached pipeline ids, and the extractor emits one `DrawBatch` per bucket with `DrawBatch::pipeline` already resolved. The 64-bit render key carries the bucket (16 layer, 8 material, 32 depth, 8 sequence), so a material batch is never split by depth. Nodes that need the mesh view-projection (`Fresnel`, `VertexPosition`) are refused in the sprite domain with a readable reason.

9. Physics: Box2D v3, joints, cloth and SdfCollisionSystem

PhysicsSystem owns one b2WorldId and exposes it to extension modules as a packed uint32, so box2d.h never leaks into a public header. Bodies are mirrored from RigidBody2D/Collider2D components into Box2D, stepped at the fixed rate, and read back before interpolation. PhysicsJointSystem builds a fixed-capacity joint graph: chain, rope and spring connections with break limits, damage accumulation, gameplay cuts, repair, in-place rebuild and ties. ClothSystem integrates a Verlet grid and projects it against the same Box2D colliders.

SdfCollisionSystem contract:

Member	Contract
SetField(const SdfField&)	Installs a non-owning mdspan over a row-major

	grid. Negative values are solid; the field's memory is owned by the renderer/terrain, not by the system.
Step(registry, physics, fixedDt)	Samples the grid at body corners and centre, converts the gradient to a normal, applies a mass-scaled push impulse, reflects inward velocity and writes the corrected state back. Returns the number of corrected bodies.
Update(SystemContext&)	ECS convenience wrapper around Step(); a headless call without physics is a no-op.
Contacts()	Fixed array of SdfContact {entity, point, normal, penetration, impulse, sequence}.
SdfCollisionDesc	pushOut, restitution, maxImpulse, subSamples, dynamicOnly.

```
Fby::SdfCollisionDesc desc;
desc.pushOut = 0.65f;
desc.restitution = 0.05f;
desc.subSamples = 2;
sdf.Initialize(desc);
sdf.SetField(terrain.Field());
sdf.Register(graph);                    // Phase::FixedUpdate

// Viewport overlay: red vectors whose length is the penetration.
debug.Build(context, Fby::Editor::PhysicsDebugSdf, &sdf);
```

> SdfCollisionSystem is not part of SystemContext. A host that wants SDF physics registers the system explicitly through Register(graph) and keeps the instance alive; the Viewport debug pass takes the same pointer.

10. NetRollbackSession: deterministic rollback

A session binds snapshot v4, CommandLog, Pcg32 and the gameplay components (tags, attributes, effects and skills) into one fixed-capacity rollback loop. Everything after Initialize() is allocation-free: one snapshot ring, one input ring per player, a fixed entity map and fixed frame metadata. The simulation callback is a function pointer plus void*.

Method	**Contract**
Initialize(registry, schema, config, arena)	Allocation-free setup with a caller-owned snapshot arena.
Initialize(registry, schema, config)	Convenience overload; allocates one arena (the only allocation the session makes).
SetSimulation(fn, user) / SetCommandLog(log)	Binds the pure simulation function and the structural command log.
Start(rng)	Captures the base tick and stores the session RNG. Required before Step().
Step(localInput)	Simulates exactly one tick with prediction for missing remote inputs and captures the snapshot.
SubmitRemoteInput(player, frame)	Stores a late or current input; a frame for an already simulated tick marks a pending rollback.
ReplayPending(report)	Restores the snapshot before the earliest corrected tick, restores RNG and CommandLog cursor, re-simulates, captures new snapshots and reports changed ticks.
CompareChecksums(tick, snap, state, out)	Compares a remote report with the stored frame; the two checksums isolate a byte-level desync from a gameplay-state desync.

```
Fby::Net::NetRollbackSession::Config cfg;
cfg.playerCount = 2;
cfg.localPlayer = 0;
cfg.rollbackTicks = 64;
session.Initialize(registry, schema, cfg, arena);
session.SetSimulation(&SimulateTick, &game);
session.Start(pcg);

while (running)
{
    session.SubmitRemoteInput(remote, received);
    session.Step(localInput);
    if (session.RollbackPending())
    {
        Fby::Net::RollbackReport report;
        session.ReplayPending(&report);
        if (report.changedTicks > 0)
        {
            LogBoy().Log(Fby::Editor::LogBoySeverity::Desync, "net",
                         "rollback {}..{} changed {}", report.fromTick, report.toTick,
                         report.changedTicks);
        }
    }
}
```

- NetInputFrame is 24 bytes of integer axes (int16 fixed point). Floats are not allowed: a float bit pattern is not guaranteed to survive a different compiler.
- Result enumerates every failure mode (NotInitialized, BufferFull, RollbackTooOld, ChecksumMismatch, ...); ResultName() is the log spelling.
- A session is non-copyable and owns its arena when the convenience overload is used.

11. Audio: mixer, DSP graph, limiter

AudioSystem owns the miniaudio device and a fixed mixer. DspGraph routes nodes into buses, supports ducking and reverb, and receives commands through a lock-free ring so the game thread never blocks the audio callback. Limiter.hpp is the final true-peak stage.

- DeafMode mutes the graph at the Master root and turns sound triggers into subtitle lines; it is a policy of the audio system, not a post-effect.
- Ogg/Opus is decoded off the audio thread; the mixer only sees cooked .fpcm buffers.

12. PinkUI and WidgetCanvasSystem

PinkUI is the in-game UI: a fixed-capacity canvas, mdspan grid layout, anchor resolution, hit testing and one-batch packing. WidgetCanvasSystem turns a WidgetCanvas into FrameOutput (draw items) without allocating; font glyphs come from a .fbyfont atlas mapped through the VFS, so the shipping runtime carries no TTF rasterizer. Glyphs are atlas-only:

there is no embedded 8x8 bitmap fallback, and a text frame without a mounted font array is refused instead of drawn with the wrong layer.

Member	Contract
WidgetCanvasSystem::MountFonts(RHI&)	Mandatory cold-start stage on the main thread. Builds the GPU texture array from the VFS .fbyfont set; a failure is a PANIC log plus FBY_ASSERT_MSG, never a fallback.
BuildFrame(context, canvas, output)	Lays out and culls one canvas, returning BuildStatus and filling FrameOutput. A text canvas without a current mounted atlas returns FontsNotMounted and writes nothing. Allocation-free; safe to call from Phase::Render.
Update(context, canvases, output)	Builds every active canvas into one output; the host submits the result as one batch. A font-contract violation stops the pass and sets fontContractViolated.
WidgetCanvas::SetDisplayScale	Density from CameraStackSystem::DisplayScale; font sizes are rounded to whole physical pixels.
OledShiftState	Integer 2 px orbit for OLED safety; ApplyOledShift() binds it to a canvas pool.

WidgetCanvasSystem is also the accessibility surface: BlindMode reads widget labels for the narrative log, DeafMode receives subtitle widgets, and colour-vision matrices plus FlashGuard are applied to the canvas colours before packing.

13. Accessibility kernel (BlindMode, DeafMode, FlashGuard)

Accessibility is a frame policy, not an overlay. BlindMode skips the game pass, indirect draws and the skinning dispatch, then publishes a bounded narrative log. DeafMode mutes the Master bus and routes sound triggers into SubtitleLines. FlashGuard compresses sudden luminance changes. Colour-vision correction uses the published Machado 2009 matrices, and SrgbRelativeLuminance()/ContrastRatio() implement WCAG 2.2 for authoring-time validation.

Item	Contract
NarrativeEvent	128-slot SPSC ring of 160-byte text payloads; the simulation thread pushes, the speech thread pops.
ColorVisionMatrix(deficiency)	Returns the 3x3 matrix for protanopia, deuteranopia or tritanopia.
ApplyColorVision(color, matrix, strength)	Applies the matrix in linear light to an sRGB Color32.
FlashGuard::Evaluate(luminance, dt)	Returns the gain for the current frame; a large delta is compressed to the threshold.
EditorSettings::ide*	The same policies applied to the editor window itself: Machado correction, WCAG high-contrast, FlashGuard, keyboard navigation and LogBoy as the narration sink.

14. Asset pipeline, VFS and localization

Assets are addressed by relative path inside the VFS. A .fempak archive is mounted with a priority; a directory mount can shadow it, which is how the editor writes a loose file before the cooker folds it back in. Read() returns a span into an mmap that stays valid until the mount is dropped or the file is hot-reloaded.

Format	Owner	Purpose
.fempak	FempakCooker	Immutable shipped archive with an index and optional encryption.
.fbyfont	FontAtlas (PinkUI)	Baked glyph atlas: [header][glyph records][pixels]; no runtime TTF rasterizer.
.femeta	AssetDatabase	Typed metadata, import settings and stable AssetId.
.fbin	LocTool	Compiled localization table; the editor UI itself is English-only.

> The editor UI is not localizable by design: menus, panel titles, errors and the inspector are English-only. Localization applies to game content and documentation, where a .fbin table and the EN/RU manual cover the shipped languages.

15. Platform services: Steam, Epic Games Store, GOG

PlatformService is the engine-side bridge; PlatformAPI.h is a flat pure-C ABI that a store SDK plugin implements. The provider is loaded dynamically, resolves only documented entry points and converts every store-specific type at the boundary, so a game never includes steam_api.h, EOSSDK or Galaxy headers.

Provider	Status in this tree	Notes
SteamProvider.cpp	Implemented	Steamworks flat API through a function table.
EgsProvider.cpp	Implemented	Epic Online Services provider through the same C ABI.
GogProvider.cpp	Implemented (shim)	Loads galaxy64.dll / libgalaxy64.so / libgalaxy64.dylib at runtime and resolves the flat fby_gog_* shim ABI; the game ships the shim built against its SDK revision. Galaxy is a C++ interface library, so a shim is mandatory. GogAchievementSystem and GogStorageSystem write the binary PlatformCommandLog in Phase::PostUpdate.
fby_fontc	Implemented	Offline TTF -> .fbyfont cooker (stb_truetype, RGBA8 atlas, verified header).

```
// Pure C ABI: the provider is a shared library, not a C++ dependency.
typedef struct FbyPlatformApi {
    uint32_t version;
    void* user;
```

```
    bool (*init)(void* user);
    void (*shutdown)(void* user);
    void (*unlockAchievement)(void* user, const char* id);
    bool (*writeCloudFile)(void* user, const char* name, const uint8_t* data, uint32_t
size);
} FbyPlatformApi;
```

16. Editor: dashboard, fonts, LogBoy, Scene Diagnostics

The dashboard is the only editor surface that exists before a project is open. It installs the editor font set before the first frame, so the first window is already Inter at the physical DPI. A new project starts in Projects/ under the engine root, resolved by DefaultProjectsRoot() - never in the process working directory and never in the install directory.

Component	Contract
EditorFonts	Loads Inter and JetBrains Mono from the VFS mmap. Subpixel advances (PixelSnapH=false), auto oversampling, no bitmap fallback, dynamic atlas baked at the viewport pixel density. Legacy backends get RasterizerDensity + inverse FontGlobalScale.
ProjectDashboard	Initialize(paths, config, visible, window, vfs, fonts, uiScale): installs typography, resolves the default Projects/ location and creates it.
LogBoySystem / LogBoyPanel	Fixed 4096-record ring of 192-byte text payloads with severity Trace/Info/Warning/Critical/Desync, FNV-1a system hashes and colour+hash filtering.
SceneDiagnosticsPanel	Per-thread occupancy bars and heatmap, FrameArena bytes, MeshRenderItem count, indirect draws, draw calls and exact simulation microseconds with avg/p99/peak.
PhysicsDebugVisualizer	Fixed-buffer line/point geometry for Box2D bodies, ropes, cloth and SDF normals, drawn over the Viewport through a caller-supplied world-to-screen map.

> The .fbyfont atlas is a baked PinkUI asset and cannot feed ImGui's vector rasterizer. The editor therefore loads the Inter TTF from the same VFS mount; the atlas header is still validated on startup, so a corrupt game-UI atlas is reported by the editor, not by the player.

17. Step-by-step: a game from an empty folder

This walkthrough produces a playable project with a sprite, physics, keyboard input, a PinkUI HUD and a deterministic rollback session. Every step maps to a real header from the index.

- 1. Launch the editor and create a project with New Project. The dashboard creates <engineRoot>/Projects/<name> with assets/{textures,audio,scenes}, configs/ and project.femproj.
- 2. Import a sprite into assets/textures. The watcher imports it, writes a .femeta and cooks a .ftex texture array layer.
- 3. Create assets/scenes/main.toml with one SpriteRenderer, one Transform2D and one RigidBody2D + Collider2D.

- 4. Register the systems: input in PreUpdate, physics in FixedUpdate, movement in Update, PinkUI in Render.
- 5. Give the camera a CameraStackSystem and call SetDisplayScale() from SDL_GetWindowDisplayScale so UI glyphs are crisp on HiDPI.
- 6. Add an AccessibilityKernel from project settings; BlindMode and DeafMode are read from project.femproj, not hard-coded.
- 7. Create a NetRollbackSession per match, bind a pure SimulateTick callback and feed NetInputFrame values.
- 8. Log through LogBoy: severity + system hash + fixed-format message. Open the LogBoy panel and filter by Desync while testing netcode.
- 9. Enable Physics Debug Draw in the Viewport while tuning colliders and cloth.
- 10. Cook with FempakCooker, run the suite with ctest and the style gate, then ship the engine-only target.

```
// 4-7 in code: systems, accessibility and rollback for one match.
Fby::Ecs::SystemGraph graph;
graph.Add(Fby::Ecs::Phase::PreUpdate, "input", &InputSystem);
graph.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem, &physics);
graph.Add(Fby::Ecs::Phase::FixedUpdate, "movement", &MovementSystem, nullptr, 1ull <<
1);
graph.Add(Fby::Ecs::Phase::Render, "ui", &UiSystem, &ui);

Fby::AccessibilityKernel kernel;
kernel.Configure(settings);              // blind/deaf/colour/flash

Fby::Net::NetRollbackSession session;
if (session.Initialize(registry, schema, cfg, arena) !=
Fby::Net::NetRollbackSession::Result::Ok)
{
    LogBoy().Log(Fby::Editor::LogBoySeverity::Critical, "net", "rollback init failed");
}
```

18. Testing, style gate and release checklist

The suite is deliberately framework-free: every test is an executable with a main() that returns non-zero on failure, so it runs on a headless CI agent, a console devkit and a developer workstation with the same command. The full suite passes at Release.

```
ctest --test-dir build --output-on-failure          # the full suite
python Source/Tools/Dev/CheckStyle.py               # style gate
python Documentation/Tools/GenerateAPIReference.py  # this manual
python Documentation/Tools/GenerateDocumentation.py
python Documentation/Tools/GenerateApplicationGuide.py
```

- A release build enables asserts and runs the whole suite once with FBY_ENABLE_ASSERTS=1, because two latent debug-only compile errors existed in JobSystem at RC.
- The POSIX branch of WriteFileAtomically() and LogLine() must be compiled on Linux/macOS in CI; the Windows-only build never saw them.
- The style gate must report zero failures before a commit is eligible for release.

19. Open work

Open work is tracked in Documentation/ROADMAP.md.

20. Appendix A - generated symbol index

Generated from the headers on every run. A symbol listed here is declared in the shipped tree; the header and line are the authoritative location.

Editor/Infrastructure (115)

Symbol	Kind	Header	Line
AIStateTreeCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	44
AnimationGraphCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	45
AnimationTimelinePanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	46
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	20
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	47
AssetImportService	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	30
AssetImportService	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	48
Callbacks	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	38
Change	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	102
Codec	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	220
Command	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	234
CommandKind	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	33
CommandStack	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	30
CommandStack	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	49
Config	Infrastructure	Source/Editor/	27

		Infrastructure/ AssetDatabaseWatchSystem.hpp	
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hpp	39
Config	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	88
ContentBrowserPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	50
CookRequest	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hpp	115
CookResult	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hpp	126
EditorCamera	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	91
EditorDesc	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	62
EditorFontRole	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	22
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	84
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/ EditorTypography.hpp	7
EditorFontSpec	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	38
EditorFontStatus	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	67
EditorSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSettings.hpp	19
EditorSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	51
EditorSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	100
Entry	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hpp	52
ExternalClientDesc	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	37
FrameComposer	RHI	Source/Engine/RHI/	64

		FrameComposer.hpp	
FrameComposerStats	RHI	Source/Engine/RHI/ FrameComposer.hpp	45
FrameGenerationMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	138
FrameGenerationPolicySetting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	131
HudTargetMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	147
ImFont	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	10
ImGuiIO	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	11
Impl	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	221
InteractionTransaction	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	16
LoadedSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	138
MaterialGraphPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	52
MaterialGraphRuntime	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime.hpp	41
MaterialRuntimeStats	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime.hpp	25
MotionHistory2D	RHI	Source/Engine/RHI/ FrameComposer.hpp	30
NetSession	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	32
NetworkDiagnosticsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	53
OitMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	98
OpaqueRenderMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	91
PendingEdit	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	255
PendingSpawn	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	264
PhysicsDebugFlags	Infrastructure	Source/Editor/	19

		Infrastructure/ PhysicsDebugDraw.hpp	
PhysicsDebugFrame	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	47
PhysicsDebugLine	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	32
PhysicsDebugPoint	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	40
PhysicsDebugVisualizer	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	58
PiePlayerInput	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	27
PlaybackIcon	Infrastructure	Source/Editor/ Infrastructure/ EditorWidgets.hpp	12
PlayInEditor	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	44
Profiler	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	27
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	57
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	254
ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	54
ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	314
ProjectDashboard	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	55
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	58
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	276
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	21
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	162
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/	22

		ReleaseCooker.hpp	
ProjectSettingsPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	56
Range	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	249
ReleaseCookConfig	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	70
ReleaseCooker	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	154
ReleaseCookProgress	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	117
ReleaseCookResult	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	138
ReleaseCookStage	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	28
ResolvedHudSettings	Infrastructure	Source/Editor/Infrastructure/ResolvedRenderSettings.hpp	35
ResolvedRenderSettings	Infrastructure	Source/Editor/Infrastructure/ResolvedRenderSettings.hpp	14
Result	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	43
Result	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	47
SdfCollisionSystem	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	13
SdfField	Physics	Source/Engine/Physics/SdfCollision.hpp	20
SdfGiVolume	Physics	Source/Engine/Physics/SdfGi.hpp	21
Services	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	220
SettingDesc	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	28
SettingKind	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	20
SettingScope	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	11
Slot	Infrastructure	Source/Editor/Infrastructure/MaterialGraphRuntim	118

		e.hpp	
Source	Infrastructure	Source/Editor/Infrastructure/EditTransactions.hpp	19
SpawnComponent	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	278
SpawnHeader	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	271
SpawnTransaction	Infrastructure	Source/Editor/Infrastructure/EditTransactions.hpp	190
StableSource	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	145
Stats	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	88
Stats	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	55
Stats	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	95
TemporalUpscaleMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	154
TextureArrayAsset	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	23
TextureBinding	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	142
TextureThumbnailCache	Infrastructure	Source/Editor/Infrastructure/TextureThumbnails.hpp	17
UpscaleMode	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	105
UpscalePolicy	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	114
UpscalerBackendSetting	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	121
UserSettings	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	59
UserSettings	Infrastructure	Source/Editor/Infrastructure/UserSettings.hpp	16
VirtualFileSystem	Infrastructure	Source/Editor/	20

		Infrastructure/AssetImportService.hpp	
VirtualFileSystem	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	15
VirtualFileSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	18
VirtualTextureSettings	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	37
VirtualTextureSettings	Infrastructure	Source/Editor/Infrastructure/ProjectConfig.hpp	31
VirtualTextureSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	22
VisualScriptPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	60
WindowManager	Infrastructure	Source/Editor/Infrastructure/EditorFonts.hpp	16

Editor/Panels (79)

Symbol	Kind	Header	Line
Action	Panels	Source/Editor/Panels/ProjectDashboard.hpp	25
AIStateTreeCanvasPanel	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	55
AIStateTreeCompiled	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	35
AIStateTreeDocument	Panels	Source/Editor/Panels/AIStateTreeCanvasPanel.hpp	24
AnimationGraphCanvasPanel	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	116
AnimationGraphDocument	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	79
AnimationGraphLink	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	65
AnimationGraphLinkKind	Panels	Source/Editor/Panels/AnimationGraphCanva	33

		sPanel.hpp	
AnimationGraphNode	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	45
AnimationGraphNodeKind	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	26
AnimationTimelinePanel	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	34
AssetImportService	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	24
AudioSystem	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	15
AudioSystem	Panels	Source/Editor/ Panels/ VoiceDevicePanel.hpp	11
CanvasModelOps	Panels	Source/Editor/ Panels/ NodeCanvasPanel.hpp	21
CommandStack	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	25
CommandStack	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	30
ComponentRegistry	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel.hpp	18
Config	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	35
ContentBrowserPanel	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	27
DragPayload	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	60
EditorFonts	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	20
EditorFonts	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	20
EditorSettingsPanel	Panels	Source/Editor/	91

		Panels/ SettingsPanels.hpp	
EditSource	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	58
EditSource	Panels	Source/Editor/ Panels/ AnimationGraphCanva sPanel.hpp	119
Entry	Panels	Source/Editor/ Panels/ ContentBrowserPanel .hpp	43
FileDialog	Panels	Source/Editor/ Panels/ ContentBrowserPanel .hpp	140
ImNodesContext	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	13
ImNodesContext	Panels	Source/Editor/ Panels/ AnimationGraphCanva sPanel.hpp	15
LevelSettingsPanel	Panels	Source/Editor/ Panels/ LevelSettingsPanel. hpp	12
LicensesPanel	Panels	Source/Editor/ Panels/ LicensesPanel.hpp	11
LogBoyPanel	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	394
LogBoyRecord	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	105
LogBoyRing	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	121
LogBoySeverity	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	33
LogBoySystem	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	186
LogicGraphDocument	Panels	Source/Editor/ Panels/ VisualScriptPanel.h pp	23
MaterialGraphPanel	Panels	Source/Editor/ Panels/ MaterialGraphPanel. hpp	20
Mode	Panels	Source/Editor/ Panels/ ProjectDashboard.hp p	84

NetSession	Panels	Source/Editor/Panels/NetworkDiagnosticsPanel.hpp	6
NetworkDiagnosticsPanel	Panels	Source/Editor/Panels/NetworkDiagnosticsPanel.hpp	15
NodeCanvasPanel	Panels	Source/Editor/Panels/NodeCanvasPanel.hpp	54
PIEAnimFrame	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	101
PieClientViewport	Panels	Source/Editor/Panels/PieClientViewport.hpp	15
PIEFrameSlot	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	289
ProjectConfigStore	Panels	Source/Editor/Panels/SettingsPanels.hpp	31
ProjectConfigStore	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	17
ProjectDashboard	Panels	Source/Editor/Panels/ProjectDashboard.hpp	22
ProjectSettingsPanel	Panels	Source/Editor/Panels/SettingsPanels.hpp	33
QueryKind	Panels	Source/Editor/Panels/LogBoyPanel.hpp	430
SceneDiagnosticsPanel	Panels	Source/Editor/Panels/SceneDiagnosticsPanel.hpp	56
SceneDiagnosticsSample	Panels	Source/Editor/Panels/SceneDiagnosticsPanel.hpp	15
ScriptDebuggerPanel	Panels	Source/Editor/Panels/ScriptDebuggerPanel.hpp	28
ScriptRuntime	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	16
SinkUser	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	88

SkeletonInspectorState	Panels	Source/Editor/Panels/SkeletonAssetPanel.hpp	19
SpawnContext	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	68
SpriteAnimationInspectorState	Panels	Source/Editor/Panels/SpriteAnimationPanel.hpp	21
SystemContext	Panels	Source/Editor/Panels/SettingsPanels.hpp	24
SystemContext	Panels	Source/Editor/Panels/SkeletonAssetPanel.hpp	13
SystemContext	Panels	Source/Editor/Panels/SpriteAnimationPanel.hpp	13
SystemName	Panels	Source/Editor/Panels/LogBoyPanel.hpp	189
TextPreviewPanel	Panels	Source/Editor/Panels/TextPreviewPanel.hpp	24
TextureThumbnailCache	Panels	Source/Editor/Panels/SpriteAnimationPanel.hpp	19
TilemapEditorPanel	Panels	Source/Editor/Panels/TilemapEditorPanel.hpp	20
TimelinePanelContext	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	22
TransportGroupPanel	Panels	Source/Editor/Panels/TransportGroupPanel.hpp	14
UserSettingsPanel	Panels	Source/Editor/Panels/UserSettingsPanel.hpp	16
VirtualFileSystem	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	18
VirtualFileSystem	Panels	Source/Editor/Panels/ProjectDashboard.hpp	13
VirtualFileSystem	Panels	Source/Editor/Panels/	15

		TextPreviewPanel.hpp	
VirtualTexturePanel	Panels	Source/Editor/Panels/VirtualTexturePanel.hpp	18
VirtualTextureSystem	Panels	Source/Editor/Panels/VirtualTexturePanel.hpp	10
VisualScriptPanel	Panels	Source/Editor/Panels/ScriptDebuggerPanel.hpp	24
VisualScriptPanel	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	32
VoiceDevicePanel	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	19
WidgetCanvas	Panels	Source/Editor/Panels/SettingsPanels.hpp	18
WindowManager	Panels	Source/Editor/Panels/ProjectDashboard.hpp	14

Engine/AI (6)

Symbol	Kind	Header	Line
AIAgentSettings	AI	Source/Engine/AI/AIController.hpp	16
AIControllerComponent	AI	Source/Engine/AI/AIController.hpp	31
AIControllerSystem	AI	Source/Engine/AI/AIController.hpp	84
AIPerceptionComponent	AI	Source/Engine/AI/AIController.hpp	59
AIPerceptionSettings	AI	Source/Engine/AI/AIController.hpp	48
AIPerceptionSystem	AI	Source/Engine/AI/AIController.hpp	67

Engine/Assets (58)

Symbol	Kind	Header	Line
AeadAlgorithm	Assets	Source/Engine/Assets/FemCrypto.hpp	18
AeadKey	Assets	Source/Engine/Assets/FemCrypto.hpp	39
AeadNonce	Assets	Source/Engine/Assets/FemCrypto.hpp	46

AeadTag	Assets	Source/Engine/Assets/FemCrypto.hpp	54
ArchiveBuildOptions	Assets	Source/Engine/Assets/FemArchiver.hpp	99
ArchiveCipher	Assets	Source/Engine/Assets/FemArchiver.hpp	29
ArchiveCodec	Assets	Source/Engine/Assets/FemArchiver.hpp	22
ArchiveEntryView	Assets	Source/Engine/Assets/FemArchiver.hpp	115
ArchiveInput	Assets	Source/Engine/Assets/FemArchiver.hpp	72
ArchiveProgress	Assets	Source/Engine/Assets/FemArchiver.hpp	85
ArchiveProgressStage	Assets	Source/Engine/Assets/FemArchiver.hpp	78
ArchiveReader	Assets	Source/Engine/Assets/FemArchiver.hpp	126
ArchiveReader	Assets	Source/Engine/Assets/FemArchiver.hpp	145
ArchiveStatus	Assets	Source/Engine/Assets/FemArchiver.hpp	36
AssetDatabase	Assets	Source/Engine/Assets/AssetDatabase.hpp	28
AssetDependency	Assets	Source/Engine/Assets/AssetMeta.hpp	83
AssetHandle	Assets	Source/Engine/Assets/AssetMeta.hpp	22
AssetImportSettings	Assets	Source/Engine/Assets/AssetMeta.hpp	45
AssetMeta	Assets	Source/Engine/Assets/AssetMeta.hpp	61
AssetRecord	Assets	Source/Engine/Assets/AssetDatabase.hpp	19
AstcBlock	Assets	Source/Engine/Assets/TextureCook.hpp	25
AtlasDesc	Assets	Source/Engine/Assets/TextureArray.hpp	31
AtlasLayer	Assets	Source/Engine/Assets/TextureArray.hpp	21

AudioClipHeader	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	87
BlockExtent	Assets	Source/Engine/ Assets/ TextureFormat.hpp	103
CleanRenderExtract	Assets	Source/Engine/ Assets/ ProjectScaffold.hpp	127
CookedAudioView	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	116
CookedTexture	Assets	Source/Engine/ Assets/ TextureCook.hpp	41
CookedTextureView	Assets	Source/Engine/ Assets/ TextureFormat.hpp	203
CookProgressFlags	Assets	Source/Engine/ Assets/ CookProgress.hpp	31
CookProgressRecord	Assets	Source/Engine/ Assets/ CookProgress.hpp	38
CookProgressStage	Assets	Source/Engine/ Assets/ CookProgress.hpp	18
CookTarget	Assets	Source/Engine/ Assets/ TextureCook.hpp	17
FanmHeader	Assets	Source/Engine/ Assets/ AnimationClipAsset. hpp	22
FemArchiver	Assets	Source/Engine/ Assets/ FemArchiver.hpp	128
FempakInput	Assets	Source/Engine/ Assets/ FempakWriter.hpp	15
FempakWriter	Assets	Source/Engine/ Assets/ FempakWriter.hpp	22
LocaleTable	Assets	Source/Engine/ Assets/ Localization.hpp	14
LocalizationGender	Assets	Source/Engine/ Assets/ Localization.hpp	21
LocalizationPack	Assets	Source/Engine/ Assets/ Localization.hpp	13
LocalizationTable	Assets	Source/Engine/ Assets/ Localization.hpp	28
MappedFile	Assets	Source/Engine/ Assets/VFS.hpp	19
MountRecord	Assets	Source/Engine/ Assets/VFS.hpp	128
PakEntry	Assets	Source/Engine/	26

		Assets/ PakFormat.hpp	
PakHeader	Assets	Source/Engine/ Assets/ PakFormat.hpp	16
PcmFormat	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	21
PinnedFile	Assets	Source/Engine/ Assets/VFS.hpp	63
ReloadNotice	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	86
StagedRow	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	108
StringEntry	Assets	Source/Engine/ Assets/ Localization.hpp	15
TextureAlphaMode	Assets	Source/Engine/ Assets/ TextureFormat.hpp	43
TextureArrayAsset	Assets	Source/Engine/ Assets/ TextureArray.hpp	51
TextureCookOptions	Assets	Source/Engine/ Assets/ TextureCook.hpp	31
TextureFormat	Assets	Source/Engine/ Assets/ TextureFormat.hpp	22
TextureHeader	Assets	Source/Engine/ Assets/ TextureFormat.hpp	184
VfsEntry	Assets	Source/Engine/ Assets/VFS.hpp	55
VirtualFileSystem	Assets	Source/Engine/ Assets/ TextureArray.hpp	16
VirtualFileSystem	Assets	Source/Engine/ Assets/VFS.hpp	69

Engine/Audio (101)

Symbol	Kind	Header	Line
AudioBus	Audio	Source/Engine/ Audio/Audio.hpp	26
AudioBusComponent	Audio	Source/Engine/ Audio/Audio.hpp	40
AudioCaptureMonitor	Audio	Source/Engine/ Audio/ AudioDevices.hpp	144
AudioClip	Audio	Source/Engine/ Audio/Audio.hpp	69
AudioCommand	Audio	Source/Engine/ Audio/DspGraph.hpp	138
AudioCommandType	Audio	Source/Engine/ Audio/DspGraph.hpp	121
AudioContextCompone	Audio	Source/Engine/	67

nt		Audio/ AudioZones.hpp	
AudioDeviceEnumerator	Audio	Source/Engine/ Audio/ AudioDevices.hpp	93
AudioDeviceInfo	Audio	Source/Engine/ Audio/ AudioDevices.hpp	54
AudioDeviceKind	Audio	Source/Engine/ Audio/ AudioDevices.hpp	15
AudioDeviceList	Audio	Source/Engine/ Audio/ AudioDevices.hpp	66
AudioDevicePreferences	Audio	Source/Engine/ Audio/ AudioDevices.hpp	73
AudioDeviceStatus	Audio	Source/Engine/ Audio/ AudioDevices.hpp	21
AudioLimiter	Audio	Source/Engine/ Audio/Limiter.hpp	23
AudioListener	Audio	Source/Engine/ Audio/Audio.hpp	94
AudioSource	Audio	Source/Engine/ Audio/Audio.hpp	79
AudioSourceComponent	Audio	Source/Engine/ Audio/Audio.hpp	53
AudioSystem	Audio	Source/Engine/ Audio/Audio.hpp	100
AudioZoneComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	32
AudioZoneKind	Audio	Source/Engine/ Audio/ AudioZones.hpp	14
AudioZoneSystem	Audio	Source/Engine/ Audio/ AudioZones.hpp	91
BiquadState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	369
BusAtomics	Audio	Source/Engine/ Audio/DspGraph.hpp	370
BusFilterParams	Audio	Source/Engine/ Audio/DspGraph.hpp	108
BusRuntime	Audio	Source/Engine/ Audio/Audio.hpp	212
BusShadow	Audio	Source/Engine/ Audio/DspGraph.hpp	291
BusState	Audio	Source/Engine/ Audio/DspGraph.hpp	338
CommandQueue	Audio	Source/Engine/ Audio/DspGraph.hpp	170
CookedClipRecord	Audio	Source/Engine/ Audio/Audio.hpp	187
DspClipView	Audio	Source/Engine/ Audio/DspGraph.hpp	205
DspGraph	Audio	Source/Engine/ Audio/DspGraph.hpp	216

DspGraph	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	16
DspResult	Audio	Source/Engine/ Audio/DspGraph.hpp	34
DspResult	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	19
DuckingParams	Audio	Source/Engine/ Audio/DspGraph.hpp	80
DuckingState	Audio	Source/Engine/ Audio/DspGraph.hpp	298
FilterState	Audio	Source/Engine/ Audio/DspGraph.hpp	327
FloatRing	Audio	Source/Engine/ Audio/Audio.hpp	24
FloatRing	Audio	Source/Engine/ Audio/ AudioStream.hpp	16
FloatRing	Audio	Source/Engine/ Audio/DspGraph.hpp	32
JitterSlot	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	119
LimiterDesc	Audio	Source/Engine/ Audio/Limiter.hpp	11
OggOpusStream	Audio	Source/Engine/ Audio/OggOpus.hpp	131
OggPacket	Audio	Source/Engine/ Audio/OggOpus.hpp	76
OggPacketReader	Audio	Source/Engine/ Audio/OggOpus.hpp	86
OggPage	Audio	Source/Engine/ Audio/OggOpus.hpp	36
OggPageReader	Audio	Source/Engine/ Audio/OggOpus.hpp	57
OpusHeadInfo	Audio	Source/Engine/ Audio/OggOpus.hpp	106
OwnedPcmRecord	Audio	Source/Engine/ Audio/Audio.hpp	194
Peer	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	129
RateBucket	Audio	Source/Engine/ Audio/ VoiceNetworkBridge. hpp	62
ResolvedZone	Audio	Source/Engine/ Audio/ AudioZones.hpp	113
ReverbParams	Audio	Source/Engine/ Audio/DspGraph.hpp	96
ReverbState	Audio	Source/Engine/ Audio/DspGraph.hpp	311
SlotRuntime	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	385
SourceRuntime	Audio	Source/Engine/ Audio/Audio.hpp	204
State	Audio	Source/Engine/	164

		Audio/ AudioDevices.hpp	
Stats	Audio	Source/Engine/ Audio/ AudioZones.hpp	94
VoiceAec	Audio	Source/Engine/ Audio/VoiceAec.hpp	44
VoiceAecSettings	Audio	Source/Engine/ Audio/VoiceAec.hpp	22
VoiceCapture	Audio	Source/Engine/ Audio/ AudioStream.hpp	64
VoiceChannel	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	65
VoiceChannelConfig	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	36
VoiceChannelStats	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	46
VoiceCodecConfig	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	15
VoiceConditionOutput	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	333
VoiceConditionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	284
VoiceConditionState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	324
VoiceDecoder	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	51
VoiceDenoiseTier	Audio	Source/Engine/ Audio/VoiceDsp.hpp	10
VoiceDistortionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	157
VoiceDspChain	Audio	Source/Engine/ Audio/VoiceDsp.hpp	99
VoiceDspSettings	Audio	Source/Engine/ Audio/VoiceDsp.hpp	43
VoiceEffectChain	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	345
VoiceEffectChainSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	221
VoiceEffectKind	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	20
VoiceEffectPresetKind	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	235
VoiceEffectProfile	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	310

VoiceEffectRack	Audio	Source/Engine/Audio/VoiceEffects.hpp	433
VoiceEffectSlot	Audio	Source/Engine/Audio/VoiceEffects.hpp	200
VoiceEncoder	Audio	Source/Engine/Audio/VoiceCodec.hpp	27
VoiceFormantSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	146
VoiceMuffleSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	80
VoiceMuteSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	67
VoiceNetworkBridge	Audio	Source/Engine/Audio/VoiceNetworkBridge.hpp	18
VoiceParticipant	Audio	Source/Engine/Audio/VoiceSession.hpp	46
VoicePitchSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	133
VoicePresetKind	Audio	Source/Engine/Audio/VoiceDsp.hpp	68
VoiceRadioSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	115
VoiceRingModulationSettings	Audio	Source/Engine/Audio/VoiceEffects.hpp	171
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InputEventType	Input	Source/Engine/ Input/ InputSystem.hpp	60
InputMappingDesc	Input	Source/Engine/ Input/ InputSystem.hpp	94
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PageSlot	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	188
PlatformKind	RHI	Source/Engine/RHI/UpscalerCapabilities.hpp	11
PostFxConstants	RHI	Source/Engine/RHI/RenderPasses.hpp	97
PostFxFlags	RHI	Source/Engine/RHI/PostProcess.hpp	10
PostFxPass	RHI	Source/Engine/RHI/RenderPasses.hpp	157
PostFxSettings	RHI	Source/Engine/RHI/PostProcess.hpp	30
RenderBlendMode	RHI	Source/Engine/RHI/RHI.hpp	190
RenderCommandRecord	RHI	Source/Engine/RHI/RenderGraph.hpp	52
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RenderGraph	RHI	Source/Engine/RHI/RenderGraph.hpp	119
RenderGraphContext	RHI	Source/Engine/RHI/RenderGraph.hpp	69
RenderGraphPass	RHI	Source/Engine/RHI/RenderGraph.hpp	106
RenderItem	RHI	Source/Engine/RHI/RenderItem.hpp	90
RenderPassDesc	RHI	Source/Engine/RHI/RenderGraph.hpp	94
RenderPassKind	RHI	Source/Engine/RHI/RenderGraph.hpp	26
RenderResourceKind	RHI	Source/Engine/RHI/RenderGraph.hpp	33
RenderTargetDesc	RHI	Source/Engine/RHI/RHI.hpp	197
RenderTargetHandle	RHI	Source/Engine/RHI/RHI.hpp	207
RHI	RHI	Source/Engine/RHI/RHI.hpp	328
SdfLightingFrame	RHI	Source/Engine/RHI/RenderItem.hpp	310
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SdfShadowConstants	RHI	Source/Engine/RHI/RenderPasses.hpp	37
SdfShadowDesc	RHI	Source/Engine/RHI/RenderPasses.hpp	18
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SequenceFooter	RHI	Source/Engine/RHI/FrameExporter.hpp	42
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ShaderPipelineDesc	RHI	Source/Engine/RHI/RHI.hpp	167
ShaderPipelineDomain	RHI	Source/Engine/RHI/RHI.hpp	161
ShaderPipelineHandle	RHI	Source/Engine/RHI/RHI.hpp	178
SkinnedVertex	RHI	Source/Engine/RHI/Mesh.hpp	30
Slot	RHI	Source/Engine/RHI/CameraStack.hpp	142
SpatialUpscaleMode	RHI	Source/Engine/RHI/RHI.hpp	90
StaticBakeCache	RHI	Source/Engine/RHI/StaticBake.hpp	58
StaticBakeRenderer	RHI	Source/Engine/RHI/StaticBakeRenderer.hpp	14
StaticBakeSettings	RHI	Source/Engine/RHI/StaticBake.hpp	10
StaticBakeStats	RHI	Source/Engine/RHI/StaticBake.hpp	47
StaticBakeTileKey	RHI	Source/Engine/RHI/StaticBake.hpp	17
TemporalResolvePass	RHI	Source/Engine/RHI/TemporalResolvePass.hpp	11

TemporalUpscaleInput	RHI	Source/Engine/RHI/TemporalUpscaler.hpp	11
TextureArrayDesc	RHI	Source/Engine/RHI/RHI.hpp	146
TextureArrayHandle	RHI	Source/Engine/RHI/RHI.hpp	304
Tile	RHI	Source/Engine/RHI/StaticBake.hpp	90
TintEntry	RHI	Source/Engine/RHI/ViewportPresenter.hpp	81
TransientRing	RHI	Source/Engine/RHI/RingBuffer.hpp	15
UiBackendInfo	RHI	Source/Engine/RHI/RHI.hpp	234
UpscalerBackend	RHI	Source/Engine/RHI/UpscalerBackend.hpp	9
UpscalerCaps	RHI	Source/Engine/RHI/RHI.hpp	98
ViewportMode	RHI	Source/Engine/RHI/ViewportModes.hpp	15
ViewportModeDescriptor	RHI	Source/Engine/RHI/ViewportModes.hpp	46
ViewportModeLegend	RHI	Source/Engine/RHI/ViewportModes.hpp	37
ViewportPresenter	RHI	Source/Engine/RHI/ViewportPresenter.hpp	29
ViewportPresenterStats	RHI	Source/Engine/RHI/ViewportPresenter.hpp	17
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VirtualFileSystem	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	16
VirtualTextureAddress	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	68
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VirtualTextureHandle	RHI	Source/Engine/RHI/RenderItem.hpp	279
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VirtualTextureRhiTarget	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	118
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VirtualTextureTileId	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	92
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Symbol	Kind	Header	Line
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AotBuildOptions	Script	Source/Engine/Script/AotBuilder.hpp	43
AotBuildResult	Script	Source/Engine/Script/AotBuilder.hpp	60
AotBuildState	Script	Source/Engine/Script/AotBuilder.hpp	19
AotGraphTranspileOptions	Script	Source/Engine/Script/AotGraphCompiler.hpp	13
AotHost	Script	Source/Engine/Script/AotHost.hpp	32
AotHost	Script	Source/Engine/Script/ScriptSystem.hpp	23
AotModule	Script	Source/Engine/Script/AotModule.hpp	22
AotProgramRef	Script	Source/Engine/Script/AotHost.hpp	22
AotTranspileOptions	Script	Source/Engine/Script/AotTranspiler.hpp	12
BoundProgram	Script	Source/Engine/Script/EcsBridge.hpp	101
CameraStackSystem	Script	Source/Engine/Script/ScriptSystem.hpp	18
CameraStackSystem	Script	Source/Engine/Script/Vm.hpp	18
Code	Script	Source/Engine/	32

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CompiledDebug	Script	Source/Engine/ Script/Bytecode.hpp	488
CompiledProgram	Script	Source/Engine/ Script/Bytecode.hpp	503
ComponentAccessor	Script	Source/Engine/ Script/ EcsBridge.hpp	17
ComponentBinding	Script	Source/Engine/ Script/Bytecode.hpp	432
Context	Script	Source/Engine/ Script/AotAbi.hpp	112
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DebugFrame	Script	Source/Engine/ Script/ ScriptDebug.hpp	103
DebugSectionHeader	Script	Source/Engine/ Script/Bytecode.hpp	181
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DebugSpanFlags	Script	Source/Engine/ Script/Bytecode.hpp	150
DebugVariable	Script	Source/Engine/ Script/Bytecode.hpp	172
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FunctionEntry	Script	Source/Engine/ Script/Bytecode.hpp	455
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LogicCompileResult	Script	Source/Engine/ Script/ GraphCompiler.hpp	748
LogicFieldBinding	Script	Source/Engine/ Script/ GraphCompiler.hpp	90
LogicGraph	Script	Source/Engine/ Script/ GraphCompiler.hpp	350
LogicLink	Script	Source/Engine/ Script/ GraphCompiler.hpp	339
LogicMathOp	Script	Source/Engine/ Script/ GraphCompiler.hpp	56
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LogicNodeKind	Script	Source/Engine/ Script/ GraphCompiler.hpp	111
Module	Script	Source/Engine/ Script/AotAbi.hpp	379
NativeId	Script	Source/Engine/ Script/Bytecode.hpp	18
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PhysicsSystem	Script	Source/Engine/ Script/ ScriptSystem.hpp	17
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ProgramHeader	Script	Source/Engine/ Script/Bytecode.hpp	465
ResolvedComponent	Script	Source/Engine/ Script/ EcsBridge.hpp	93
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ScriptEvent	Script	Source/Engine/ Script/Bytecode.hpp	205
ScriptRuntime	Script	Source/Engine/ Script/ ScriptSystem.hpp	22
ScriptRuntime	Script	Source/Engine/ Script/ ScriptSystem.hpp	26
ScriptVm	Script	Source/Engine/ Script/Vm.hpp	115
ScriptVm	Script	Source/Engine/ Script/Vm.hpp	229
Slot	Script	Source/Engine/ Script/AotHost.hpp	94
Slot	Script	Source/Engine/ Script/ ScriptSystem.hpp	140
SourceLocation	Script	Source/Engine/ Script/ ScriptDebug.hpp	17
TextCompileOptions	Script	Source/Engine/ Script/ TextScript.hpp	14
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VmFault	Script	Source/Engine/ Script/Vm.hpp	30
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Engine/Text (68)

Symbol	Kind	Header	Line
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ChainedShaper	Text	Source/Engine/ Text/TextShaper.hpp	133
CodepointShaper	Text	Source/Engine/ Text/TextShaper.hpp	99
Entry	Text	Source/Engine/ Text/TextShaper.hpp	247
FontCandidate	Text	Source/Engine/ Text/ TextFontStack.hpp	24
FontSlot	Text	Source/Engine/ Text/ TextFontStack.hpp	59
FormatEntry	Text	Source/Engine/ Text/ TextService.hpp	583
FormattedLayoutEntry	Text	Source/Engine/ Text/ TextService.hpp	566
FormattedLayoutKey	Text	Source/Engine/ Text/ TextService.hpp	556
HarfBuzzShaper	Text	Source/Engine/ Text/TextShaper.hpp	110
Impl	Text	Source/Engine/ Text/TextShaper.hpp	128
ITextShaper	Text	Source/Engine/ Text/TextShaper.hpp	89
LayoutEntry	Text	Source/Engine/ Text/ TextService.hpp	548
LayoutKey	Text	Source/Engine/ Text/ TextService.hpp	528
MappedTextAsset	Text	Source/Engine/ Text/ TextAssetFile.hpp	17
MarkupStackEntry	Text	Source/Engine/ Text/TextMarkup.hpp	71
MarkupVariantEntry	Text	Source/Engine/ Text/ TextService.hpp	574
ScriptBackend	Text	Source/Engine/ Text/ TextShapeSystem.hpp	111

ScriptShaper	Text	Source/Engine/Text/TextShaper.hpp	207
ShapedGlyph	Text	Source/Engine/Text/TextShaper.hpp	56
ShapedGlyphHit	Text	Source/Engine/Text/TextShapedLayout.hpp	153
ShapedLayoutResult	Text	Source/Engine/Text/TextShapedLayout.hpp	26
ShapedLine	Text	Source/Engine/Text/TextShapedLayout.hpp	18
StyledAsset	Text	Source/Engine/Text/TextService.hpp	595
TextAction	Text	Source/Engine/Text/TextStyle.hpp	61
TextActionKind	Text	Source/Engine/Text/TextStyle.hpp	51
TextAnimationParams	Text	Source/Engine/Text/TextRender.hpp	29
TextAssetBlock	Text	Source/Engine/Text/TextAssetFormat.hpp	34
TextAssetHeader	Text	Source/Engine/Text/TextAssetFormat.hpp	20
TextAssetRef	Text	Source/Engine/Text/TextAsset.hpp	14
TextAssetStatus	Text	Source/Engine/Text/TextAssetFormat.hpp	63
TextAssetView	Text	Source/Engine/Text/TextAssetFormat.hpp	42
TextBlock	Text	Source/Engine/Text/TextAsset.hpp	37
TextDirection	Text	Source/Engine/Text/TextShaper.hpp	21
TextFontStack	Text	Source/Engine/Text/TextFontStack.hpp	54
TextFormatArg	Text	Source/Engine/Text/TextFormat.hpp	45
TextFormatStatus	Text	Source/Engine/Text/TextFormat.hpp	33
TextGradientParams	Text	Source/Engine/Text/TextRender.hpp	22
TextInlineHit	Text	Source/Engine/Text/TextLayout.hpp	204
TextInlineKind	Text	Source/Engine/Text/TextStyle.hpp	73
TextInlineObject	Text	Source/Engine/Text/TextStyle.hpp	83
TextInlineSprite	Text	Source/Engine/Text/TextStyle.hpp	99

TextLayoutParams	Text	Source/Engine/Text/TextLayout.hpp	54
TextLayoutResult	Text	Source/Engine/Text/TextLayout.hpp	72
TextLine	Text	Source/Engine/Text/TextLayout.hpp	22
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TextRenderEffects	Text	Source/Engine/Text/TextRender.hpp	44
TextRenderParams	Text	Source/Engine/Text/TextRender.hpp	57
TextRenderStats	Text	Source/Engine/Text/TextRender.hpp	76
TextRenderSystemStats	Text	Source/Engine/Text/TextRenderSystem.hpp	19
TextRun	Text	Source/Engine/Text/TextShaper.hpp	47
TextRuntime	Text	Source/Engine/Text/TextRuntime.hpp	17
TextScript	Text	Source/Engine/Text/TextShaper.hpp	30
TextService	Text	Source/Engine/Text/TextService.hpp	18
TextShapeCache	Text	Source/Engine/Text/TextShapeCache.hpp	59
TextShapeHandle	Text	Source/Engine/Text/TextShapeCache.hpp	19
TextShapeKey	Text	Source/Engine/Text/TextShapeCache.hpp	32
TextShapeParams	Text	Source/Engine/Text/TextShaper.hpp	71
TextShapeResult	Text	Source/Engine/Text/TextShaper.hpp	82
TextShapeSystemStats	Text	Source/Engine/Text/TextShapeSystem.hpp	16
TextShapeView	Text	Source/Engine/Text/TextShapeCache.hpp	50
TextSpan	Text	Source/Engine/Text/TextStyle.hpp	40
TextStore	Text	Source/Engine/Text/TextStore.hpp	19
TextStyle	Text	Source/Engine/Text/TextStyle.hpp	27
TextStyledBlock	Text	Source/Engine/Text/TextMarkup.hpp	30
TextStyleFlags	Text	Source/Engine/Text/TextStyle.hpp	12
UnicodeBoundary	Text	Source/Engine/Text/UnicodeBreak.hpp	13

VerticalFormEntry	Text	Source/Engine/Text/TextVertical.hpp	16

Engine/UI (101)

Symbol	Kind	Header	Line
AccessibilityEditorSettings	UI	Source/Engine/UI/Accessibility.hpp	250
AccessibilityFlags	UI	Source/Engine/UI/Accessibility.hpp	208
AccessibilityKernel	UI	Source/Engine/UI/Accessibility.hpp	304
AccessibilityOverlay	UI	Source/Engine/UI/AccessibilityOverlay.hpp	30
AccessibilityPresentationSettings	UI	Source/Engine/UI/Accessibility.hpp	233
AccessibilityProfile	UI	Source/Engine/UI/Accessibility.hpp	272
AccessibilitySettings	UI	Source/Engine/UI/Accessibility.hpp	217
AudioSystem	UI	Source/Engine/UI/Accessibility.hpp	26
AvSpeechTts	UI	Source/Engine/UI/Tts.hpp	60
BuildStatus	UI	Source/Engine/UI/PinkUI.hpp	1440
CardBoardComponent	UI	Source/Engine/UI/CardBoard.hpp	49
CardBoardSystem	UI	Source/Engine/UI/CardBoard.hpp	74
ColorMatrix3	UI	Source/Engine/UI/Accessibility.hpp	42
ColorVisionDeficiency	UI	Source/Engine/UI/Accessibility.hpp	34
Dropdown	UI	Source/Engine/UI/PinkUI.hpp	534
FlashGuard	UI	Source/Engine/UI/Accessibility.hpp	59
FocusDirection	UI	Source/Engine/UI/PinkUI.hpp	211
FocusPolicy	UI	Source/Engine/UI/PinkUI.hpp	187
FontAtlas	UI	Source/Engine/UI/FontAtlas.hpp	101
FontAtlasHeader	UI	Source/Engine/UI/FontAtlas.hpp	45
FontAtlasSet	UI	Source/Engine/UI/FontAtlas.hpp	167
FontBuildOptions	UI	Source/Engine/UI/FontAtlas.hpp	25
FontBuildOptions	UI	Source/Engine/UI/FontBuilder.hpp	15
FontCoverage	UI	Source/Engine/UI/FontCoverage.hpp	21
FontFamilyFile	UI	Source/Engine/UI/FontAtlas.hpp	74
FontFamilyManifest	UI	Source/Engine/UI/	17

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FontPixelFormat	UI	Source/Engine/UI/ FontAtlas.hpp	37
FontStyle	UI	Source/Engine/UI/ FontAtlas.hpp	65
FrameOutput	UI	Source/Engine/UI/ PinkUI.hpp	1418
GlyphRecord	UI	Source/Engine/UI/ FontAtlas.hpp	80
GridView	UI	Source/Engine/UI/ PinkUI.hpp	424
HyperlinkRange	UI	Source/Engine/UI/ Hypertext.hpp	51
HypertextComponent	UI	Source/Engine/UI/ Hypertext.hpp	32
HypertextFlags	UI	Source/Engine/UI/ Hypertext.hpp	21
HypertextPool	UI	Source/Engine/UI/ Hypertext.hpp	74
HypertextView	UI	Source/Engine/UI/ Hypertext.hpp	67
ITextShaper	UI	Source/Engine/UI/ PinkUI.hpp	33
LayoutCache	UI	Source/Engine/UI/ PinkUI.hpp	736
LazyFontFamily	UI	Source/Engine/UI/ FontFamilyAsset.hpp	101
ListView	UI	Source/Engine/UI/ PinkUI.hpp	560
LocalizationTable	UI	Source/Engine/UI/ Accessibility.hpp	24
LocalizationTable	UI	Source/Engine/UI/ Hypertext.hpp	16
LocalizationTable	UI	Source/Engine/UI/ PinkUI.hpp	39
NarrativeArchetype	UI	Source/Engine/UI/ NarrativeTree.hpp	23
NarrativeEvent	UI	Source/Engine/UI/ Accessibility.hpp	83
NarrativeGroup	UI	Source/Engine/UI/ NarrativeTree.hpp	37
NarrativeLocale	UI	Source/Engine/UI/ NarrativeTree.hpp	17
NarrativeLog	UI	Source/Engine/UI/ Accessibility.hpp	99
NarrativeNode	UI	Source/Engine/UI/ NarrativeTree.hpp	47
NarrativeTree	UI	Source/Engine/UI/ NarrativeTree.hpp	55
OledShiftState	UI	Source/Engine/UI/ PinkUI.hpp	463
PinkSdfDraw	UI	Source/Engine/UI/ PinkSdf.hpp	74
PinkSdfFlags	UI	Source/Engine/UI/ PinkSdf.hpp	24
PinkSdfGpuPrimitive	UI	Source/Engine/UI/ PinkSdf.hpp	51
PinkSdfPrimitive	UI	Source/Engine/UI/ PinkSdf.hpp	33

PinkSdfShape	UI	Source/Engine/UI/PinkSdf.hpp	15
RadialProgress	UI	Source/Engine/UI/PinkUI.hpp	584
RetainedFont	UI	Source/Engine/UI/FontAtlas.hpp	293
SapiTts	UI	Source/Engine/UI/Tts.hpp	30
ScrollView	UI	Source/Engine/UI/PinkUI.hpp	491
SlotComponent	UI	Source/Engine/UI/CardBoard.hpp	27
SlotFlags	UI	Source/Engine/UI/CardBoard.hpp	15
SlotLayout	UI	Source/Engine/UI/CardBoard.hpp	65
SoundCueCategory	UI	Source/Engine/UI/SoundCues.hpp	26
SoundCueDef	UI	Source/Engine/UI/SoundCues.hpp	62
SoundCueRegistry	UI	Source/Engine/UI/SoundCues.hpp	71
SoundTrigger	UI	Source/Engine/UI/Accessibility.hpp	128
StateTrack	UI	Source/Engine/UI/Accessibility.hpp	417
SubtitleLine	UI	Source/Engine/UI/Accessibility.hpp	139
SubtitleQueue	UI	Source/Engine/UI/Accessibility.hpp	163
SystemContext	UI	Source/Engine/UI/PinkUI.hpp	28
TabBar	UI	Source/Engine/UI/PinkUI.hpp	515
TextAlign	UI	Source/Engine/UI/PinkUI.hpp	222
TextShapeParams	UI	Source/Engine/UI/PinkUI.hpp	34
TextVerticalAlign	UI	Source/Engine/UI/PinkUI.hpp	229
TimedSoundCue	UI	Source/Engine/UI/Accessibility.hpp	152
TooltipPlacer	UI	Source/Engine/UI/PinkUI.hpp	595
TtsBackend	UI	Source/Engine/UI/Accessibility.hpp	27
TtsBackend	UI	Source/Engine/UI/Tts.hpp	19
TtsPriority	UI	Source/Engine/UI/Tts.hpp	11
UiPalette	UI	Source/Engine/UI/UiTheme.hpp	41
UiPaletteSlot	UI	Source/Engine/UI/UiTheme.hpp	18
UiThemeAsset	UI	Source/Engine/UI/UiTheme.hpp	46
VirtualFileSystem	UI	Source/Engine/UI/FontAtlas.hpp	19
VoiceCaptionEvent	UI	Source/Engine/UI/	193

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VoiceCaptionProvider	UI	Source/Engine/UI/Accessibility.hpp	201
WidgetCanvas	UI	Source/Engine/UI/Accessibility.hpp	30
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	290
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	483
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	607
WidgetCanvasSystem	UI	Source/Engine/UI/PinkUI.hpp	1453
WidgetComponent	UI	Source/Engine/UI/PinkUI.hpp	251
WidgetComponentFlags	UI	Source/Engine/UI/PinkUI.hpp	263
WidgetDesc	UI	Source/Engine/UI/PinkUI.hpp	309
WidgetFlags	UI	Source/Engine/UI/PinkUI.hpp	145
WidgetHandle	UI	Source/Engine/UI/PinkUI.hpp	81
WidgetHyperlink	UI	Source/Engine/UI/PinkUI.hpp	300
WidgetKind	UI	Source/Engine/UI/PinkUI.hpp	99
WidgetLayoutAsset	UI	Source/Engine/UI/WidgetLayoutAsset.hpp	15
WidgetPaintContext	UI	Source/Engine/UI/PinkUI.hpp	291
WidgetPaintContext	UI	Source/Engine/UI/PinkUI.hpp	391

Engine/Video (20)

Symbol	Kind	Header	Line
AssetDatabase	Video	Source/Engine/Video/VideoStream.hpp	23
ClusterScan	Video	Source/Engine/Video/WebmDemuxer.hpp	108
ContainerKind	Video	Source/Engine/Video/WebmDemuxer.hpp	150
DecodeStatus	Video	Source/Engine/Video/VideoTypes.hpp	29
DemuxStatus	Video	Source/Engine/Video/WebmDemuxer.hpp	52
Element	Video	Source/Engine/Video/WebmDemuxer.hpp	85
FrameBuffer	Video	Source/Engine/Video/VideoPlayer.hpp	118

FrameSlot	Video	Source/Engine/Video/VideoPlayer.hpp	125
MappedFile	Video	Source/Engine/Video/VideoPlayer.hpp	22
RHI	Video	Source/Engine/Video/VideoPlayer.hpp	27
VideoPlayer	Video	Source/Engine/Video/VideoPlayer.hpp	33
VideoStreamComponent	Video	Source/Engine/Video/VideoStream.hpp	36
VideoStreamFlags	Video	Source/Engine/Video/VideoTypes.hpp	20
VideoSystem	Video	Source/Engine/Video/VideoStream.hpp	49
VirtualFileSystem	Video	Source/Engine/Video/VideoStream.hpp	20
Vp9Backend	Video	Source/Engine/Video/Vp9Backend.hpp	30
WebmDemuxer	Video	Source/Engine/Video/WebmDemuxer.hpp	60
WebmFrame	Video	Source/Engine/Video/WebmDemuxer.hpp	38
WebmTrackInfo	Video	Source/Engine/Video/WebmDemuxer.hpp	14
YuvFrameView	Video	Source/Engine/Video/Vp9Backend.hpp	12

Engine/Weather (12)

Symbol	Kind	Header	Line
ClothSystem	Weather	Source/Engine/Weather/Weather.hpp	13
WeatherAtmospherePass	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	128
WeatherContextComponent	Weather	Source/Engine/Weather/Weather.hpp	32
WeatherDesc	Weather	Source/Engine/Weather/Weather.hpp	91
WeatherFlags	Weather	Source/Engine/Weather/Weather.hpp	19
WeatherGpuConstants	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	46

WeatherGpuResources	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	18
WeatherGpuStats	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	68
WeatherPreconciliatedDesc	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	32
WeatherPreconciliatedSystem	Weather	Source/Engine/Weather/WeatherPreconciliated.hpp	78
WeatherSystem	Weather	Source/Engine/Weather/Weather.hpp	99
WeatherWindField	Weather	Source/Engine/Weather/Weather.hpp	75

21. Appendix B - macros and platform defines

Macro	Header	Line
FBY_AOT_EXPORT	Source/Engine/Script/AotAbi.hpp	396
FBY_AOT_EXPORT	Source/Engine/Script/AotAbi.hpp	398
FBY_ASSERT	Source/Engine/Core/Assert.hpp	13
FBY_ASSERT	Source/Engine/Core/Assert.hpp	31
FBY_ASSERT_MSG	Source/Engine/Core/Assert.hpp	22
FBY_ASSERT_MSG	Source/Engine/Core/Assert.hpp	32
FBY_DEFINE_PLUGIN	Source/Engine/Core/Plugin.hpp	56
FBY_DETAIL_CONCAT	Source/Engine/Core/Definitions.hpp	261
FBY_DETAIL_CONCAT_INNER	Source/Engine/Core/Definitions.hpp	260
FBY_FIELD	Source/Engine/Core/Mirror.hpp	284
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/Fluid.hpp	16
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/Fluid.hpp	18
FBY_HAS_MDSPAN	Source/Engine/ECS/Skeleton.hpp	17
FBY_HAS_MDSPAN	Source/Engine/ECS/Skeleton.hpp	19
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/Mirror.hpp	102
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/Mirror.hpp	105
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/	109

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FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/FlatMap.hpp	19
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/FlatMap.hpp	21
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/FlatMap.hpp	24
FBY_PLATFORM_ABI_VERSION	Source/Plugins/PlatformService/PlatformAPI.h	11
FBY_PLATFORM_CALL	Source/Plugins/PlatformService/PlatformAPI.h	15
FBY_PLATFORM_CALL	Source/Plugins/PlatformService/PlatformAPI.h	18
FBY_PLATFORM_EXPORT	Source/Plugins/PlatformService/PlatformAPI.h	14
FBY_PLATFORM_EXPORT	Source/Plugins/PlatformService/PlatformAPI.h	17
FBY_PLUGIN_ABI_VERSION	Source/Engine/Core/PluginAPI.h	28
FBY_PLUGIN_API_H	Source/Engine/Core/PluginAPI.h	18
FBY_PLUGIN_COMPONENT	Source/Engine/Core/Plugin.hpp	53
FBY_PLUGIN_ENTRY	Source/Engine/Core/PluginAPI.h	115
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginAPI.h	31
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginAPI.h	33
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginAPI.h	35
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